

Reusable learning supports a larger architecture design system

Reusable architecture knowledge

Prior projects and literature

design artifacts + documents
tool traces + measurements
failures + reviewed results

Current project record

Version-controlled study state

workload + software versions
constraints + candidate identity
tool, library + process versions
current measurements + failures

Architecture role

supplies intent + constraints
reviews result + recommends
architecture commitment

AI-assisted architecture design system

Architecture foundation model

optional reusable learned component
adapted to the current task

Architecture tasks

workload + requirements analysis

prediction + design-space screening

candidate, software + RTL generation

diagnosis + verification assistance

Conventional methods

models, algorithms, solvers

Design tools

compilers + simulators
RTL, EDA + formal tools

Measurements and checks

qualified tool returns + failures
assumptions + validity limits
independent rejection conditions

Supported architecture result

comparison + measurements + assumptions + checks + limits

The moonshot is complete-system capability, not autonomous prompt-to-chip generation.